ScapeNet

Natural scene classification with deep learning

A pretrained model compared with a CNN trained from scratch

Deep Learning semester project

Problem and dataset

6 scene classes

buildings, forest, glacier, mountain, sea and street

Kaggle dataset

Intel Image Classification, a secondary dataset of 150 × 150 colour photos

14,000 + 3,000

About 14,000 photos for training and validation, 3,000 kept for testing

Two models, one dataset

Property	Pretrained MobileNetV2	Our CNN (final model)
Starting weights	ImageNet pretrained	Random, no pretraining
Training	16 epochs, 2 stages	38 epochs, 2 runs
Test accuracy	90.4%	84.1%
Test loss	0.2533	0.4644
Role in the project	Test model for comparison	Final model we trained

Results on 3,000 test photos

90.4%

Pretrained MobileNetV2

84.1%

Our CNN, trained from scratch

Class (F1)	Pretrained	Ours
buildings	0.90	0.81
forest	0.99	0.97
glacier	0.85	0.78
mountain	0.85	0.78
sea	0.93	0.85
street	0.92	0.86

Problems we faced and fixed

Where	Problem	Fix
Kaggle	No kaggle.json button	Used legacy API credentials
Colab	Browser converter crashed	Exported to ONNX instead
Colab	Our CNN was unstable	Trained again, lower learning rate
Hugging Face	CPU hardware needs a PRO plan	Used free ZeroGPU
Hugging Face	Missing keras, GPU task aborted	Added requirements.txt, loaded model inside the GPU function
Hugging Face	Visitors exceeded the free GPU limit	Moved both models into the browser
Colab	ONNX export failed	Removed training-only layers, checked outputs match

How the live demo works

1

Upload

The user picks a photo on the ScapeNet website

2

Load

The page loads the chosen model (ONNX) into the browser

3

Predict

The model runs on the visitor's device and returns six probabilities

4

Show

Top class and a bar for every class, for one model or both

Conclusion

- Pretraining helps: 90.4% against 84.1% on the same 3,000 photos.
- Our own CNN works well, and both models run in the browser for anyone to test.
- Glacier and mountain are the hardest pair. More data and stronger augmentation come next.

ScapeNet: code, report and live demo are on the project website and GitHub.